A multi-epitope vaccine against the highly pathogenic SARS-CoV-2

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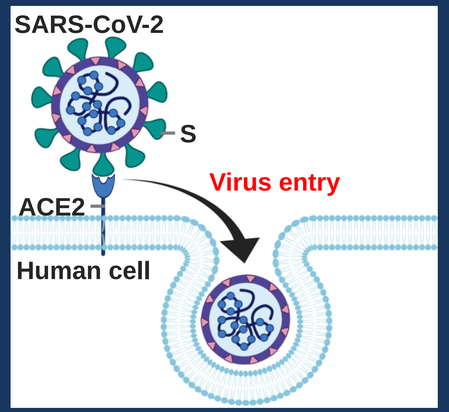
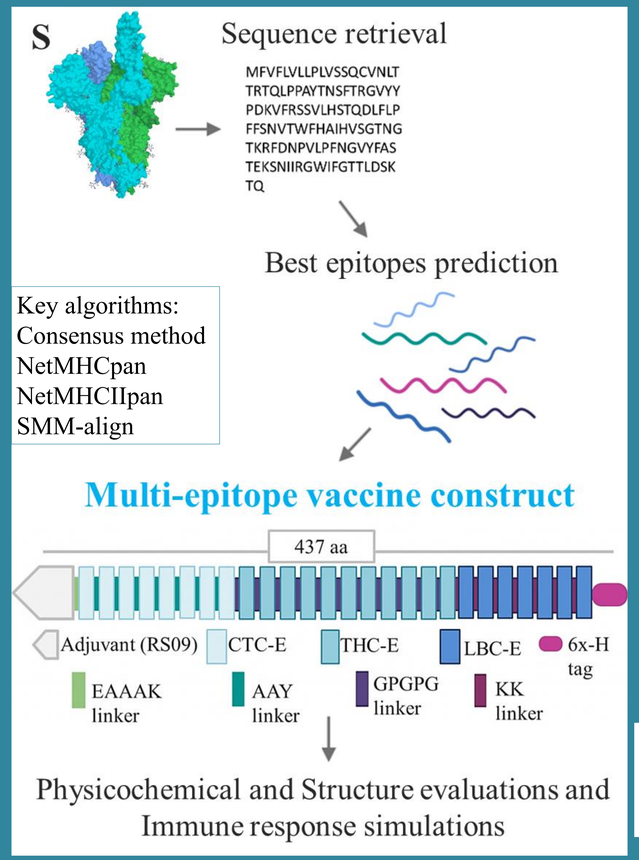
Introduction

COVID-19 represents a new global threat demanding a multidisciplinary effort to fight its etiological agent: **SARS-CoV-2**. Immunoinformatics may aid to predict prominent immunogenic regions from critical SARS-CoV-2 proteins for their use in prophylactic and/or therapeutic interventions. This work aimed to design a **high-quality vaccine** against SARS-CoV-2 using the Spike (**S**) glycoprotein as a target, which mediates virus entry into human cells:

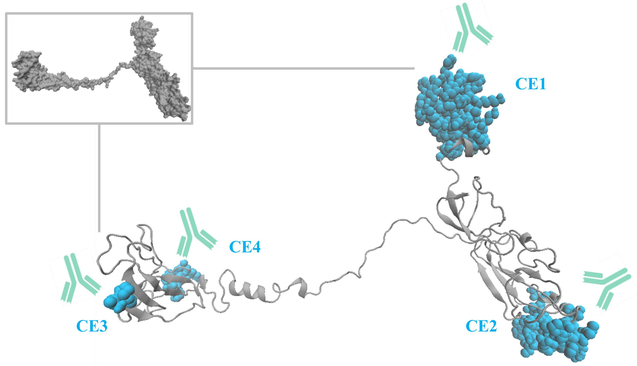


Methods

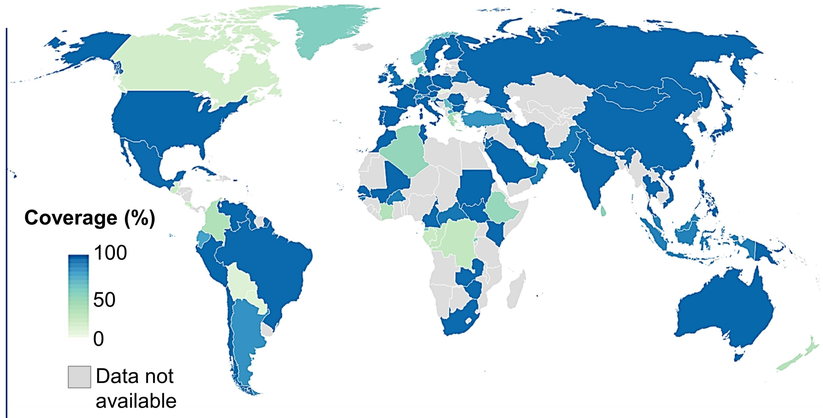
The vaccine model was design using an Integrated *in silico* approach:



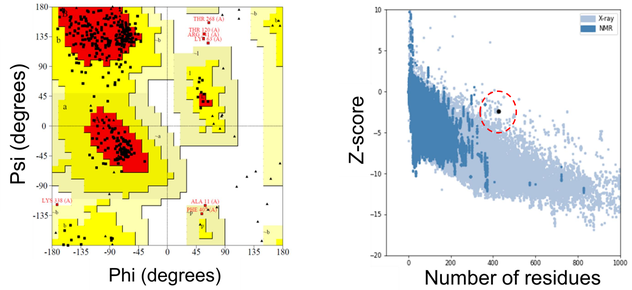
Results



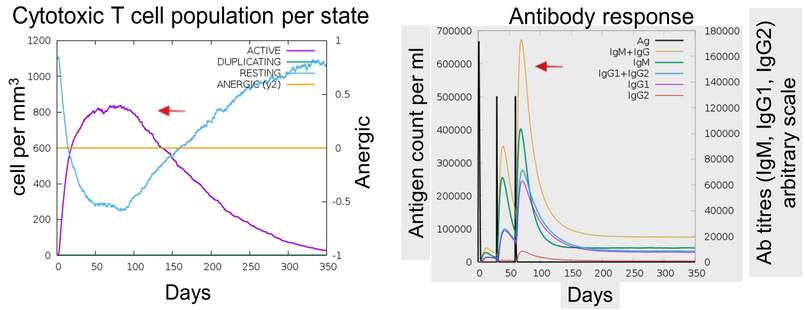
Refined 3D structure of the vaccine model. Space-filling model (inset). Epitopes (cyan atoms) for antibodies (represent in turquoise).



Global population coverage of the vaccine model.



Validation studies of the vaccine model. Ramachandran plot (left) and Z-score (right).



Immune response simulations (3 doses given 4 weeks apart).

Conclusions

- Notably, computer-based simulations suggest that the vaccine model has a robust potential to evoke and maximize both immune effector responses and immunological memory to SARS-CoV-2.
- Further research is needed to accomplish with the international guidelines for human vaccine formulations.